
SIGN-AWARE ANTITHETIC COUPLING FOR SIGNED GRADIENT-PARTICLE METHODS

PREPRINT

Stephen Abkin
Department of Mathematics
Texas A&M University
College Station, TX, USA
stephen122204@tamu.edu

Prabir Daripa
Department of Mathematics
Texas A&M University
College Station, TX, USA
daripa@tamu.edu

September 10, 2026

ABSTRACT

Signed gradient-particle methods reconstruct a field by cumulatively summing particle masses. Their sampling error depends on both particle positions and mass signs, which makes the coupling of repeated simulations consequential. We study an antithetic estimator that matches two replicas by spatial rank and uses reflected Gaussian increments for like-sign masses and synchronous increments for unlike-sign masses. The construction specializes established coupling principles to transient field estimation without changing either replica’s discrete law. We prove an exact nonnegative reduction in conditional integrated variance when the switch is applied at the final diffusion stage. The identity identifies the relevant mass products and separations; it does not imply a variance ordering for the complete nonlinear evolution. For a two-pulse viscous Burgers problem, the proposed estimator has integrated variance 0.739 times that of within-sign pairing, with an approximate 95% bootstrap interval $[0.682, 0.801]$, and integrated mean squared error 0.819 times as large. Its additional benefit is observed across fixed profiles, observation times, and a cubic-flux test. In a held-out comparison, selected plans using four and six trajectories respectively attain the same expected-error target; the proposed estimator takes approximately 0.60 of the execution time of within-sign pairing. Heat and one-sign controls, nonlinear observables, and bias-dominated regimes delimit these conclusions.

Keywords gradient random walk · signed particles · antithetic coupling · variance reduction · viscous conservation laws · Monte Carlo

1 Introduction

Gradient-particle methods approximate the spatial derivative of a solution by a signed particle measure and recover the field by cumulative summation. This representation concentrates particles near steep gradients and supports a mesh-free field reconstruction. Gradient random walks have been developed for reaction–diffusion and viscous conservation equations [1–5]. More recently, Bertaglia et al. [6] extended gradient-based Monte Carlo constructions to relaxation approximations of hyperbolic conservation laws, including systems. The present work concerns repeated estimation of a transient field within an existing one-dimensional signed-particle solver.

The computational question is how to couple the randomness in two replicas. Independent averaging reduces sampling variance but requires additional trajectories. Antithetic sampling instead seeks a negative covariance between the outputs of two runs with unchanged marginal laws [7, 8]. In a signed cumulative reconstruction, opposite masses reverse the response to a particle displacement. Reflection of all matched Gaussian increments therefore need not give a favorable field covariance, even when the particles are close in space.

We match particles by spatial rank and reflect their increments when the matched masses have the same sign. When their signs differ, we use synchronous increments. A natural alternative also uses the signs: pair particles within each

sign class and reflect every matched increment. The comparison between these two sign-aware rules is central. It separates the benefit of retaining spatial matching from the benefit already obtained by respecting mass signs.

1.1 Related particle and coupling methods

The coupling principles have established antecedents. Nüsken and Pavliotis [9] formulate variance reduction through couplings with preserved marginals. Their Proposition 54 and Corollaries 57–58 include reflection/synchronization switches based on products of response-related quantities; Section 5.2 describes repeated sorting and pairwise matching within a collection of sampler copies. Their asymptotic variance analysis motivates such constructions, but does not provide a finite-time variance comparison for the signed interacting solver studied here. The choice of spatial rank and mass sign is our specific realization of these ideas for a cumulative field estimator. It requires no auxiliary equation, although computable switching criteria also occur in the cited special cases.

Within the gradient-particle literature, Lécot [10] combines signed cumulative reconstruction, spatial ordering, and quasi-random diffusion inputs for a Burgers construction. A related treatment appears in Lécot and Ogawa [11]. This is the closest representation-specific precedent for structuring the random inputs. Our randomized quasi-Monte Carlo comparison adapts that diffusion strategy to the transport update used here; it is not a reproduction of the complete published algorithm.

Recent QMC particle constructions also couple noise within an interacting ensemble. Ben Rached et al. [12] prove weak convergence for additive-noise McKean–Vlasov dynamics and develop an antithetic multilevel QMC estimator. Their construction and regularity assumptions differ from our signed rank-dependent solver. Together with array-RQMC work for a Fokker–Planck particle model [13], this places the diffusion adaptation below within a broader literature; its result cannot rank QMC methods as a family.

Bencheikh and Jourdain [14] study bias and antithetic sampling for mean-field particle approximations. Their equation (3.1) couples a $2N$ -particle system with two N -particle systems to form a particle-number correction, an object different from our same-size replica average. Related convergence and bias analyses [15–19] concern particular underlying particle schemes and assumptions. Preservation of our baseline’s law does not extend those theorems to a different reconstruction or transport convention.

Variance reduction can also act through a control variate, conditional expectation, or a modified particle representation [8, 20–22]. We compare the coupling with two known-mean controls and conditional evaluation of the last diffusion stage. These comparisons concern the selected estimators and their implementations, rather than the best attainable performance of each variance-reduction family.

1.2 Contributions and organization

The contribution has three components. First, we specify a sign-aware spatial-rank coupling for signed cumulative field estimation and establish preservation of each replica’s complete discrete law. This retains the baseline expectation and allows its discretization bias to be studied separately from the covariance introduced by pairing. Second, we derive an exact final-stage variance identity and identify an unconditional equality case when all masses have one sign. The identity is local to the conditioned diffusion stage; a counterexample shows why an unrestricted coordinatewise-monotonicity argument cannot supply the missing nonlinear terminal comparison. Third, we quantify the additional computational benefit over within-sign pairing, alongside reflected sampling, controls, and a randomized quasi-Monte Carlo adaptation. Fixed-profile experiments, two distinct held-out target comparisons, and nonlinear observables establish the empirical scope.

The intended use is ensemble field estimation after a signed-particle solver has been selected. We do not compare its efficiency with deterministic PDE solvers. The numerical evidence is one-dimensional and scalar, and concerns the mean-transport update defined below. In particular it does not establish an advantage for sampled two-speed transport.

Section 2 defines the update and estimators. Section 3 gives their exact properties. Section 4 presents the comparisons and applicability tests, followed by the discussion and conclusion. Appendices supply proofs, reference and initialization details, and the statistical methodology.

2 The solver, the estimators and the coupling

2.1 Problem class and signed particle representation

We consider the scalar viscous conservation law

$$\partial_t u + \partial_x f(u) = \nu \partial_{xx} u, \quad u(x, 0) = u_0(x), \quad x \in \mathbb{R}, \quad 0 < t \leq T. \quad (2.1)$$

We take $f \in C^1(\mathbb{R})$ and $\nu > 0$, with initial data u_0 of bounded variation whose derivative is integrable and whose limits $u_{\pm} = \lim_{x \rightarrow \pm\infty} u_0(x)$ exist. The experiments use $f(u) = u^2/2$ (viscous Burgers), $f(u) = u^3/3$, and $f \equiv 0$ (heat), the last as a control in which the coupling acts without transport.

Differentiating (2.1) gives an equation for $w = \partial_x u$, which the method represents by a signed atomic measure carried by N particles,

$$w^N(\cdot, t) = \sum_{i=1}^N m_i \delta_{X_i(t)}, \quad U^N(x, t) = u_- + \sum_{i: X_i(t) \leq x} m_i. \quad (2.2)$$

The masses m_i are fixed at initialisation and never change; only the positions evolve. For the equal-far-field pulse profiles, we initialise deterministically, placing $N/2$ particles at midpoint quantiles of the positive part of u'_0 and $N/2$ at those of the negative part, each with $|m_i| = \text{TV}(u_0)/N$ (Appendix C). This gives equal counts and equal absolute mass per sign class, and $\sum_i m_i = u_+ - u_-$; for the smooth pulse profiles $u_+ = u_-$, so the total signed mass is zero. The one-sign control instead has nonzero total signed mass and is initialized separately in Appendix C. Zero total signed mass is a hypothesis of Lemma 3.4 below and we flag where it is used.

2.2 One step of a single replica

Fix a step $h > 0$, assume $K = T/h$ is an integer, and write $\sigma = \sqrt{2\nu h}$. A replica keeps a state (X, m, v) of positions, masses and carried velocities. Its storage order is the most recent pre-diffusion rank order; after diffusion the positions need not be sorted. Algorithm 1 is one step: transport with the velocity carried from the previous step, re-sort, read the new velocity off the reconstructed field at the particle (inclusive of the particle's own mass), and then diffuse.

Algorithm 1: One step of a single replica (mean-transport gradient random walk)

Input: state (X, m, v) with associated entries, step h , viscosity ν , flux derivative f'

- 1 $X_i \leftarrow X_i + v_i h$ // transport with the carried velocity;
 - 2 $\pi \leftarrow$ a stable sort of X ; $(X, m) \leftarrow (X_\pi, m_\pi)$;
 - 3 $v_i \leftarrow f'(u_- + \sum_{j \leq i} m_j)$; // new carried velocity, inclusive cumulative sum
 - 4 draw $Z \sim \mathcal{N}(0, I_N)$ independent of the past; $X_i \leftarrow X_i + \sigma Z_i$;
-

Three features of Algorithm 1 matter later. The velocity is computed *after* the sort and *before* the diffusion, and is used by the *next* transport, so it is part of the state and must stay associated with its particle until used. At the subsequent sort the old velocity is discarded and recomputed; if a state is reordered without recomputation, its velocity must be permuted too. The sort is stable, so ties are broken by prior order. And the particles are not re-sorted between the diffusion and the following transport; the next sort does that.

The velocity is evaluated from the rank-inclusive cumulative sum. At tied positions this convention uses the stable particle order; it is not the value of a field reconstruction that includes all co-located masses at once. The same convention is used in every coupling. For Burgers, this is the conditional-mean variant of the underlying two-speed implementation; here it serves as a fixed baseline.

2.3 Two replicas, a matching, and a multiplier

Both replicas start from the same deterministic initial state. They are advanced in lockstep. At stage k let

$$S_k = (X^A, m^A, v^A, X^B, m^B, v^B)$$

be the joint state *after* the sort and the velocity update of Algorithm 1 and *before* the diffusion, and let $\mathcal{F}_k$ contain the complete joint history through this pre-diffusion stage, including all previous random draws.

Definition 2.1 (admissible coupling). An *admissible coupling* is a rule that, at each stage k , selects a bijection $\pi_k : \{1, \dots, N\} \rightarrow \{1, \dots, N\}$ and multipliers $\varepsilon_k \in \{-1, +1\}^N$, both $\mathcal{F}_k$ -measurable, draws $Z^{(k)} \sim \mathcal{N}(0, I_N)$ independent of $\mathcal{F}_k$ and of all earlier draws, and sets

$$X_i^A \leftarrow X_i^A + \sigma Z_i^{(k)}, \quad X_j^B \leftarrow X_j^B + \sigma \varepsilon_{k,j} Z_{\pi_k(j)}^{(k)}. \quad (2.3)$$

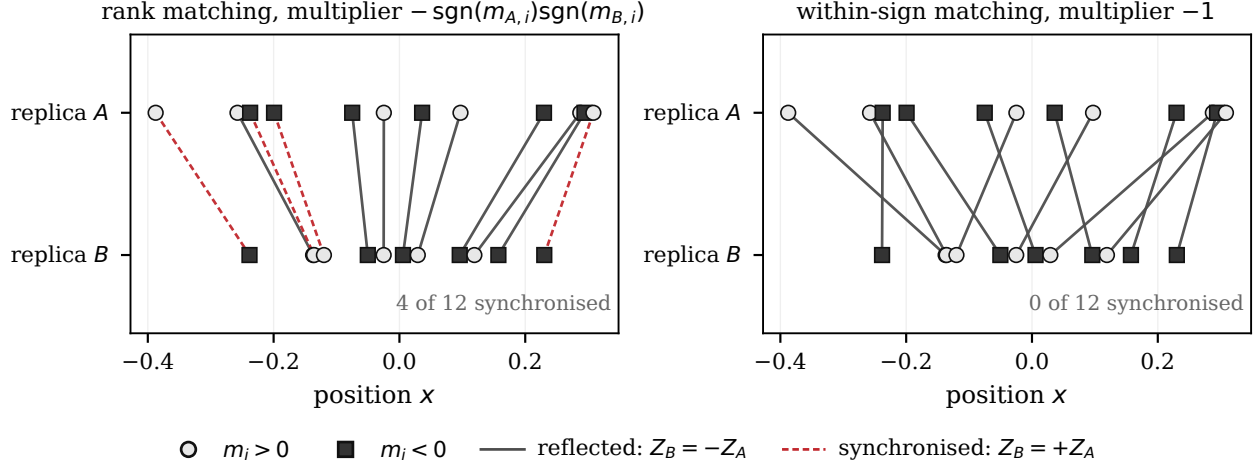


Figure 1: The two matchings on the same paired state, for the interleaved-sign configuration just before the tenth diffusion stage with $N = 12$ (regenerated by `analysis/make_figures.py`; state archived as `analysis/fig1_state.npz`). Left: rank matching pairs spatially particles of equal rank; unlike-sign pairs are synchronised rather than reflected. Right: within-sign matching never produces an unlike-sign pair, but its partners are further apart. Both rules are bijections measurable with respect to the pre-diffusion joint state, so both preserve each replica’s marginal law (Proposition 3.1).

2.4 Estimators, error functional and work

Write $U^{(A)}, U^{(B)}$ for the two replicas’ reconstructed fields (2.2) at time T . Three estimators are compared:

$$\text{single run: } \hat{u}_B^{\text{sing}} = \frac{1}{B} \sum_{b=1}^B U^{(b)}, \quad B \text{ independent runs}; \quad (2.5)$$

$$\text{one pair: } \bar{U}^{(b)} = \frac{1}{2} (U^{(A,b)} + U^{(B,b)}); \quad (2.6)$$

$$\text{ensemble of pairs: } \hat{u}_B = \frac{1}{B} \sum_{b=1}^B \bar{U}^{(b)}, \quad B \text{ independent replicates of the pair.} \quad (2.7)$$

Accuracy is measured by the expected integrated squared error over a window $\Omega = [-5, 5]$ discretised by $L = 400$ equispaced points x_ℓ with spacing Δx :

$$\mathcal{E}(B) = \mathbb{E} \|\hat{u}_B - u_{\text{ref}}\|_\Omega^2, \quad \|g\|_\Omega^2 = \Delta x \sum_{\ell=1}^L g(x_\ell)^2. \quad (2.8)$$

Because the replicates are independent and identically distributed,

$$\mathcal{E}(B) = b^2 + \frac{V}{B}, \quad b^2 = \|\mathbb{E} \bar{U} - u_{\text{ref}}\|_\Omega^2, \quad V = \Delta x \sum_{\ell} \text{Var}(\bar{U}(x_\ell)). \quad (2.9)$$

We call V the *integrated variance* and b^2 the *squared bias*; b^2 contains both the finite- N representation error and the time-stepping error, as well as any reference approximation error. Here u_{ref} is an independently computed spectral or Cole–Hopf solution (Appendix C). Work is measured as $W = B \cdot C$ with C the wall-clock cost of one replicate, timed end to end under the timing protocols specified for each experiment; tuning costs, where a plan was selected from a pilot, are reported separately and never folded into C .

Two remarks on (2.8). First, Ω is a bounded window, whereas the identity of Lemma 3.4 and Proposition 3.5 concerns the integral over all of $\mathbb{R}$; the two are kept apart in what follows. Second, for a nonlinear functional Q the two natural estimators $Q(\bar{U})$ and $\frac{1}{2}(Q(U^{(A)}) + Q(U^{(B)}))$ have different expectations, so a variance ratio is not an accuracy ratio; §4.5 reports empirical mean squared error for those.

3 Properties of the coupled estimator

Law preservation applies to the full discrete trajectory. The variance identity below instead conditions on a single paired state immediately before diffusion. This distinction is necessary because earlier coupling choices change the state on which later stages act.

3.1 Each replica keeps its law

Proposition 3.1 (marginal law preservation). *Let two replicas be advanced by Algorithm 1 from the same initial state under any admissible coupling (Definition 2.1). Then the law of the complete state trajectory $(X_k^A, m_k^A, v_k^A)_{k=0}^K$ of replica A, and likewise of replica B, is exactly that of an uncoupled single replica. Only the joint law of the two changes.*

The proof (Appendix A) is an induction: a predictable permutation of an i.i.d. Gaussian vector is i.i.d. Gaussian, and predictable ± 1 multipliers preserve $\mathcal{N}(0, \sigma^2)$ coordinatewise, so each replica’s one-step transition kernel is unchanged given the past.

Remark 3.2 (what the hypotheses are, and what they are not). The proposition requires the conditional Gaussian transition and replica-local deterministic update specified above. In particular it does *not* need the particle signs to be constant in time, nor the output functional to be linear in the particle measure. Constant signs are a property of this initialisation, not a requirement for law preservation; a scheme in which masses were updated would still preserve each replica’s law provided the updates are functions of that replica’s own state and the multipliers remain predictable and of unit modulus.

Corollary 3.3 (shared bias). *Under any admissible coupling, $\mathbb{E}\bar{U} = \mathbb{E}U^{\text{sing}}$ pointwise, provided the displayed expectations exist. Thus the squared bias in (2.9) is identical for REFLECT, SWITCH, and WITHIN under the same baseline initialization and update.*

The conclusion follows by linearity of expectation over the two replicas. More generally it holds for the average of any integrable per-replica functional. It does not hold automatically for a nonlinear function of the pair mean. Preservation of the baseline law retains its own valid expectation and bias results, but it neither establishes continuum convergence nor imports error rates proved for other particle methods.

3.2 An exact identity for the final diffusion stage

The following elementary identity is what lets a variance of the reconstructed field be written in terms of particle separations. It is a whole-line statement and it requires zero total signed mass.

Lemma 3.4 (whole-line product identity). *Let $D_A(x) = \sum_i m_i^A \mathbf{1}\{X_i^A \leq x\}$ and $D_B(x) = \sum_j m_j^B \mathbf{1}\{X_j^B \leq x\}$ with $\sum_i m_i^A = \sum_j m_j^B = 0$. Then D_A, D_B are compactly supported and*

$$\int_{\mathbb{R}} D_A(x) D_B(x) dx = -\frac{1}{2} \sum_{i,j} m_i^A m_j^B |X_i^A - X_j^B|. \quad (3.1)$$

Proposition 3.5 (final-stage identity). *Let $g_s(d) = \mathbb{E}|d + sZ|$ with $Z \sim \mathcal{N}(0, 1)$, and condition on the paired state immediately before the last diffusion stage, with a_i, b_i the positions of the rank- i particles of A and B, $d_i = a_i - b_i$, and masses m_i^A, m_i^B satisfying the hypothesis of Lemma 3.4. Write $V_{\bullet} = \int_{\mathbb{R}} \text{Var}(U(x) \mid \text{state}) dx$ for the conditional whole-line integrated variance of the pair mean under coupling $\bullet$. Then*

$$V_{\text{REFLECT}} - V_{\text{SWITCH}} = \frac{1}{4} \sum_{i: m_i^A m_i^B < 0} |m_i^A m_i^B| \left[g_{2\sigma}(d_i) - |d_i| \right] \geq 0. \quad (3.2)$$

Only matched pairs of unlike mass contribute; each contributes in proportion to the product of the mass magnitudes and to the excess of $g_{2\sigma}(d_i)$ over $|d_i|$, which is largest when the matched separation is small relative to σ and decays as $|d_i|/\sigma$ grows. Non-negativity is Jensen’s inequality. The proof (Appendix A) uses Lemma 3.4 twice and the observation that all off-diagonal terms $i \neq j$ are unaffected by the coupling, because the two increments involved are then independent under the two rank-matched rules being compared.

Scope, and how much of the effect it explains. *One stage.* Equation (3.2) does not say that switching helps whenever the matched signs disagree at an earlier stage; the state distributions at later stages are themselves coupling-dependent. We checked the identity against a direct paired Monte Carlo over the final stage at six pre-final states drawn from the

two-pulse problem; all six agreed within a nominal 95% interval built from 12 000 draws in 40 batches, and refining the quadrature window by a factor of 2.7 moved the estimate by 0.02%. Averaging the right-hand side over reflected histories estimates the reduction obtained by switching only their last stage: the conditional mean is unchanged, so the law of total variance gives this comparison exactly. Dividing that estimated reduction by the separately measured full-history reduction gives 3.75%, 3.71%, and 1.95% in three archived Gaussian/two-pulse cells. These are descriptive ratios of estimates, not a guaranteed fraction of the full-history gain. The identity therefore explains a local improvement but leaves most of the measured terminal comparison to numerical investigation.

3.3 When the two couplings are identical

Lemma 3.6 (conditional identity). *Start both couplings from the same initial state and drive them with the same sequence of draws $Z^{(k)}$. Then their complete states coincide at every stage through the pre-diffusion state of the first stage at which a matched pair has unlike masses; their diffusion updates at that stage can differ. In particular, if no such stage occurs before K , the two runs are identical.*

The proof is immediate: while $\text{sgn}(m_i^A) \text{sgn}(m_i^B) = +1$ for all i , the multiplier (2.4) equals -1 , which is REFLECT.

Corollary 3.7 (one-sign case). *If every gradient mass has the same sign, the hypothesis of Lemma 3.6 holds for every history, and SWITCH and REFLECT are the same algorithm.*

Corollary 3.7 is a genuine limitation, not a corner case: for monotone initial data the sign switch makes no change to reflected pairing. Reflected pairing itself can still reduce variance relative to independent runs. We verified it on the full state – positions, masses and carried velocities – for 200 seeds under both heat and Burgers dynamics, with maximum absolute difference exactly 0 in every seed.

Spatial separation is not enough. It is tempting to expect the same conclusion when the positive and negative regions start far apart, since then the rank order groups the particles by sign. This is false, and the reason is elementary: Gaussian support is unbounded, so for two particles separated by d the probability that exactly one of the antithetic replicas reverses their order is $\Pr(|Z_1 - Z_2| > d/\sigma) > 0$ at every finite d . Once an order reverses, a matched pair of unlike masses appears and the two couplings part company. On our separated configuration at $N = 2048$ with 200 fresh seeds, the two couplings diverge in $58/200 = 0.290$ [0.228, 0.358] of seeds under heat dynamics and $59/200 = 0.295$ [0.233, 0.363] under Burgers (exact binomial intervals), with median first divergence at step 187 of 200.

3.4 No unrestricted monotonicity

Remark 3.8 (coordinatewise monotonicity fails). Coordinatewise monotonicity of a response is a standard sufficient condition for antithetic variance reduction [8]. The corresponding orientation of the terminal field in each particle’s initial position fails for Algorithm 1. With masses $(1/2, -1/2)$, $h = 0.005$ and $\nu = 10^{-10}$, moving the positive particle right from -0.00275 to -0.00225 , while keeping the negative particle at zero, increases the expected field at $x = 0.00125$ after two steps. Appendix B gives the exact zero-noise trace and a Gaussian-tail bound establishing the strict inequality for positive viscosity.

This counterexample excludes the unrestricted monotonicity argument for the specified discrete update. It does not compare the terminal variances of SWITCH and REFLECT, and does not exclude a result with resolution assumptions. Its transport-to-diffusion ratio $|m|h/\sigma = 2500$ lies far outside the main experiments.

3.5 A geometric fact about rank matching

One quantity reported in §4.3 is the mean matched-pair separation under each matching rule. Its interpretation depends on an elementary fact that we state here so that it is not mistaken for evidence.

Remark 3.9 (sorted matching is the optimal matching for absolute distance). For two equally sized finite sets of reals $\{a_i\}$ and $\{b_j\}$ listed in increasing order, the identity bijection minimises $\sum_i |a_i - b_{\pi(i)}|$ over all bijections π ; this is the monotone rearrangement, optimal for costs of the form $c(a - b)$ with convex c [23]. Rank matching is that bijection, and within-sign matching is a bijection constrained to respect the sign classes. Hence

$$\sum_i |a_i - b_i| \leq \sum_i |a_i - b_{\pi_{\text{WITHIN}}(i)}|$$

with probability one, and the ratio of mean matched separations reported in Table 4 is bounded above by 1 by construction.

What that ratio measures is therefore *how much geometric slack the sign-class constraint costs* in a given configuration, not whether separation drives the variance difference. Proposition 3.5 makes the final-stage gain depend on matched separation relative to σ , which is why the quantity is worth reporting at all; but a number that cannot exceed 1 cannot by itself confirm a mechanism, and §4.3 treats it accordingly.

4 Numerical study

4.1 Setup, and how uncertainty is reported

Unless stated otherwise the test problem is the asymmetric signed two-pulse initial condition

$$u_0(x) = 0.8 e^{-(x+0.7)^2/(2 \cdot 0.5^2)} - 0.5 e^{-(x-0.4)^2/(2 \cdot 0.3^2)}, \quad (4.1)$$

with $f(u) = u^2/2$, $\nu = 0.1$, $T = 1$ and $h = 0.005$ ($K = 200$ steps), evaluated on $\Omega = [-5, 5]$ at $L = 400$ points. It has three sign regions in u'_0 , so both sign classes are populated and the classes interleave once. Reference solutions are Fourier pseudospectral on $[-16, 16]$ with $M = 2048$ modes, $\frac{2}{3}$ dealiasing and an integrating-factor RK4 step of 10^{-4} ; for Burgers this agrees with a stabilised whole-line Cole–Hopf evaluation to 5.6×10^{-14} in the maximum norm, and for the cubic flux it self-converges to 1.6×10^{-13} (Appendix C).

All reported experiments use recorded deterministic seed keys; independent replicates use distinct streams. The archive specifies the complete key tuple for each experiment, including whether the comparison arms share a stream. Whole-replicate resampling preserves that dependence when present. The main field, control and nonlinear-MSE tables use 6000 percentile-bootstrap resamples; the profile/time and structure tables retain their archived 3000 and 2000 resamples, respectively. Scalar station variance ratios use $F(n-1, n-1)$ intervals for independent arms. Held-out errors use t intervals on 32 independent block errors, with Bonferroni correction across the arms of each experiment. Divergence frequencies use exact binomial intervals. The intervals are approximate except where an exact model-based construction is specified. Appendix D gives an empirical scalar calibration and its limits; it does not validate every field or MSE interval.

Reading a ratio. Throughout, a ratio below 1 favours the numerator and an interval containing 1 means the comparison is *unresolved at this sample size*. It does not mean the effect is absent, and we never report it as such.

4.2 Comparison with alternative estimators

The comparison that decides whether the construction is worth its complexity is not against reflected pairing. It is against within-sign pairing, which is what a reader who has noticed the sign problem would write, and against the variance reduction a reader would reach for instead of changing the coupling at all.

The headline comparison. Table 1 gives the three pairwise comparisons at the production cell, from an independent replay of the archived experiment. Being sign-aware at all is most of the gain: within-sign pairing already reaches 0.172 of reflected variance. Rank matching with a switched multiplier reaches 0.127, a further factor of 1.35, and the interval for that further factor excludes 1.

Table 1: Integrated variance and integrated mean squared error over $\Omega = [-5, 5]$ for the pair mean, two-pulse Burgers, $N = 8192$, $h = 0.005$, 48 replicates; whole-replicate percentile bootstrap intervals. Absolute values: integrated variance 4.326×10^{-5} (reflected), 7.433×10^{-6} (within-sign), 5.492×10^{-6} (sign-switched).

comparison	integrated variance ratio	integrated MSE ratio
within-sign / reflected	0.1718 [0.1450, 0.2038]	0.2056 [0.1738, 0.2443]
sign-switched / reflected	0.1269 [0.1073, 0.1511]	0.1684 [0.1425, 0.2007]
sign-switched / within-sign	0.7389 [0.6817, 0.8006]	0.8193 [0.7542, 0.8928]

Figure 2 illustrates cancellation of the two replica errors in one paired realization. Its variance panel, computed over all 48 replicates, shows that the improvement varies substantially with location. These are spatial diagnostics of this field comparison, not independent confirmations or pointwise performance guarantees. Section 4.5 gives separate station comparisons with uncertainty.

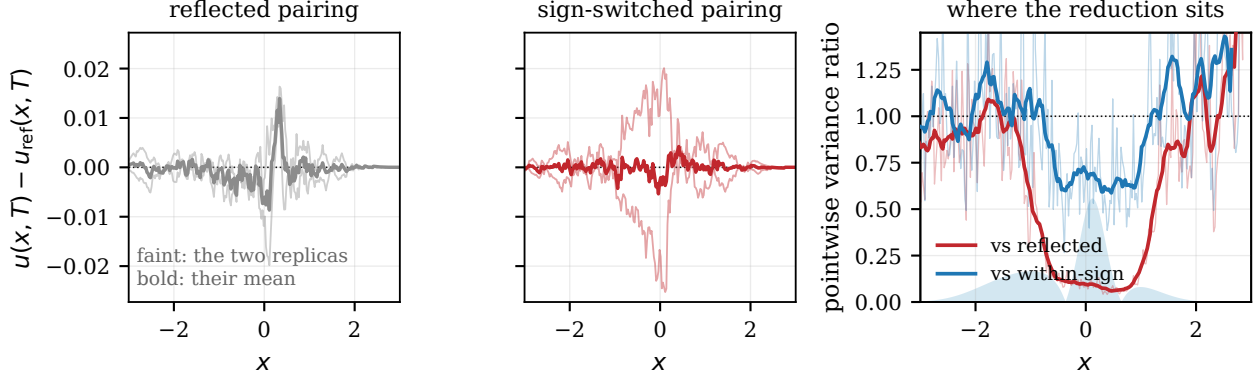


Figure 2: Two-pulse Burgers, $N = 8192$, 48 replicates, from the independently replayed archive. Left and centre: the pointwise error of each replica (faint) and of their mean (bold) on one replicate. Right: the ratio of pointwise variance of the sign-switched pair mean to that of the reflected (red) and within-sign (blue) pair means; thin lines are the raw ratios over 48 replicates and thick lines a 13-point running mean. The shaded profile is $|\partial_x u_{\text{ref}}|$, scaled to the panel.

Cost. A pair costs 1.93 times a single trajectory at $N = 8192$ (0.1950 s against 0.1009 s, serial, from experiment A). In a separate interleaved timing batch, among the three pairings, at $N = 8192$ one replicate costs 0.1941 s (reflected), 0.1995 s (sign-switched) and 0.2117 s (within-sign): the switch costs 1.028 times reflected pairing and 0.943 times within-sign pairing, and at $N = 2048$ the corresponding figures are 1.040 and 0.908. Combining the measured cost with Table 1 gives a variance $\times$ cost ratio of 0.697 against within-sign pairing. **These are implementation results.** In these experiments the sign classes always have equal counts in both replicas, so the handling of unequal counts cannot be what makes within-sign pairing slower here; partitioning and gathering are the plausible causes, but we did not profile, and we therefore do not claim an intrinsic cost advantage over all within-sign implementations. Timings are serial and single-machine.

Against variance reduction that does not change the coupling. Table 2 compares the couplings with four alternatives applied to the same solver: conditional averaging of the final diffusion stage (RB), which replaces the terminal field by its exact conditional expectation; a known-mean heat control (CVH); and a known-mean control built from a frozen initial velocity attached to each particle identity (CVF), each with unit coefficient and with a coefficient fitted on a disjoint training batch. All arms share seed streams, so the intervals are paired. For a pre-final configuration (a_i, m_i) , this conditional field is

$$u_- + \sum_i m_i \Phi((x - a_i)/\sigma),$$

averaged over the two replicas. It integrates the indicator rather than setting the displacement to zero. The heat and frozen-velocity controls use each particle identity's accumulated Brownian displacement W_i and terminal surrogates $X_i^0 + W_i$ and $X_i^0 + v_i^0 T + W_i$, respectively. Their exact means are

$$\mu_H(x) = u_- + \sum_i m_i \Phi\left(\frac{x - X_i^0}{\sqrt{2\nu T}}\right), \quad \mu_F(x) = u_- + \sum_i m_i \Phi\left(\frac{x - X_i^0 - v_i^0 T}{\sqrt{2\nu T}}\right).$$

Here v_i^0 is the deterministic initial rank velocity, carried by identity in the surrogate. Predictable sign/permutation transforms preserve the conditional Gaussian law of each identity's fresh increment, hence its accumulated increment has law $\mathcal{N}(0, 2\nu T)$. This proves the displayed means for the discrete initialization. A control estimator is $\bar{U} - c(\bar{H} - \mu_H)$ or its frozen-velocity counterpart. The fitted scalar c minimizes integrated variance on a disjoint training sample by dividing integrated sample covariance by integrated control variance; it is then held fixed for evaluation.

Three things in Table 2 matter. First, the best control-variate arm built on reflected pairing reaches variance 0.672 but variance $\times$ cost 0.923 on Burgers, before its 0.47 s of coefficient training: the sign switch is more effective than this control addition at the reported cell. Second, once the coupling is switched, a scalar control adds essentially nothing on Burgers ($0.1649 \rightarrow 0.1642$) and little on the cubic flux ($0.1739 \rightarrow 0.1684$, a resolved but modest 0.968); this is a statement about these two controls at these configurations and not a claim that no control can help. Third, conditional averaging of the final stage is *complementary* rather than competing: applied on top of the switched coupling it halves the variance again and gives the best variance $\times$ cost in the table (0.151). Its work benefit depends on the cost of evaluating the conditional CDF sum: in a grid-sensitivity test the additional variance–cost ratio changes from 0.614 at

Table 2: Alternatives on the same solver: integrated variance relative to reflected pairing, with paired whole-replicate bootstrap intervals, at $N = 1600$, $h = 0.005$, 64 seeds. “ $v \times c$ ” multiplies the variance ratio by the measured per-replicate cost ratio; one-off coefficient training, where required, is listed separately and is not amortised in $v \times c$. c^* denotes a coefficient fitted on a disjoint batch and frozen.

flux	arm	variance / reflected	$v \times c$	one-off (s)	variance / within-sign
Burgers	reflected	1.0000	1.000	–	–
	+ conditional final stage	0.8813 [0.852, 0.909]	1.612	–	–
	+ heat control, c^*	0.8440 [0.821, 0.869]	1.170	0.488	–
	+ frozen-velocity control, c^*	0.6724 [0.617, 0.740]	0.923	0.466	–
	within-sign	0.2083 [0.179, 0.245]	0.243	–	1
	sign-switched	0.1649 [0.142, 0.194]	0.174	–	0.7916 [0.762, 0.825]
	+ frozen-velocity control, c^*	0.1642 [0.141, 0.194]	0.237	0.490	0.7886 [0.758, 0.822]
cubic	+ conditional final stage	0.0804 [0.069, 0.095]	0.151	–	0.3861 [0.362, 0.412]
	reflected	1.0000	1.000	–	–
	+ frozen-velocity control, c^*	0.5031 [0.458, 0.552]	0.697	0.448	–
	within-sign	0.2263 [0.196, 0.262]	0.268	–	1
	sign-switched	0.1739 [0.151, 0.201]	0.184	–	0.7688 [0.741, 0.797]
	+ heat control, c^*	0.1684 [0.146, 0.194]	0.248	0.496	0.7442 [0.716, 0.773]
	+ conditional final stage	0.0818 [0.070, 0.096]	0.158	–	0.3614 [0.340, 0.383]

201 probes to 1.485 at 1601 probes. Thus its variance guarantee does not guarantee lower computational work. We keep the two separate throughout so that the coupling’s contribution is never inflated by conditioning.

A fourth alternative, a randomised quasi-Monte Carlo adaptation of [Lécot’s](#) idea applied to our transport solver, is included in the held-out experiment in §4.4.1; it attained the target with 46 trajectories against four.

4.3 Dependence on profile, time, and flux

We next vary the initial profile, observation time, and flux to test the scope of the fixed-setting comparison.

Profiles and times. Three fixed initial profiles have different sign structures: a single bump (2 regions), the two-pulse profile (4.1) (3), and an oscillatory profile $0.45 \sin(3x)e^{-x^2/(2 \cdot 1.1^2)}$ (13). Each is observed at $T = 0.25, 1.0$ and 2.0 with $N = 2048$ and 48 replicates. Table 3 and the left panel of Figure 3 give the result.

Table 3: Integrated variance ratio of sign-switched to within-sign pairing, $N = 2048$, $h = 0.005$, 48 replicates, whole-replicate bootstrap intervals.

profile	sign regions	$T = 0.25$	$T = 1.0$	$T = 2.0$
single bump	2	0.958 [0.886, 1.039]	0.790 [0.727, 0.862]	0.810 [0.735, 0.893]
two-pulse	3	0.895 [0.830, 0.967]	0.785 [0.713, 0.866]	0.793 [0.702, 0.889]
oscillatory	13	0.932 [0.876, 0.990]	0.667 [0.613, 0.725]	0.595 [0.544, 0.655]

Eight of the nine cells resolve an advantage. The exception is the single bump at the earliest time, 0.958 [0.886, 1.039], which is unresolved – meaning no resolved advantage at 48 replicates, not that the benefit disappears. The advantage is larger at $T = 1$ and $T = 2$ than at $T = 0.25$ in every profile, and is largest for the oscillatory profile at the two later times. It is **not** monotone in time: the single bump moves from 0.790 to 0.810 and the two-pulse profile from 0.785 to 0.793 between $T = 1$ and $T = 2$. The supported statement is that the advantage is stronger at the later times than at the earliest time in these examples.

What this does and does not isolate. The three profiles differ in amplitude, shape and gradient magnitude as well as in the number of sign regions, so interleaving is *not* isolated as the cause of the difference between them. We report the ordering as an observation, not a demonstration.

Table 4 reports the associated geometry. Proposition 3.5 makes the final-stage gain depend on matched separation relative to σ , and the sign-class constraint does cost geometric slack, most at the oscillatory profile and at later times. But Remark 3.9 shows that the direction of this table is fixed in advance: rank matching is the optimal matching for absolute distance, so the ratio is at most 1 whatever the configuration. The table quantifies the size of the slack; it cannot

Table 4: Mean matched-pair separation under rank matching divided by that under within-sign matching, both rules applied to the *same* paired state so the history is held fixed; $N = 2048$, 24 replicates, ± 2 standard errors. By Remark 3.9 every entry is at most 1 with probability one; what varies is the size of the gap.

profile	sign regions	$T = 0.25$	$T = 1.0$	$T = 2.0$
single bump	2	0.928 [0.908, 0.948]	0.851 [0.817, 0.885]	0.790 [0.741, 0.840]
two-pulse	3	0.897 [0.879, 0.916]	0.839 [0.803, 0.875]	0.820 [0.762, 0.878]
oscillatory	13	0.847 [0.832, 0.862]	0.693 [0.655, 0.732]	0.676 [0.624, 0.728]

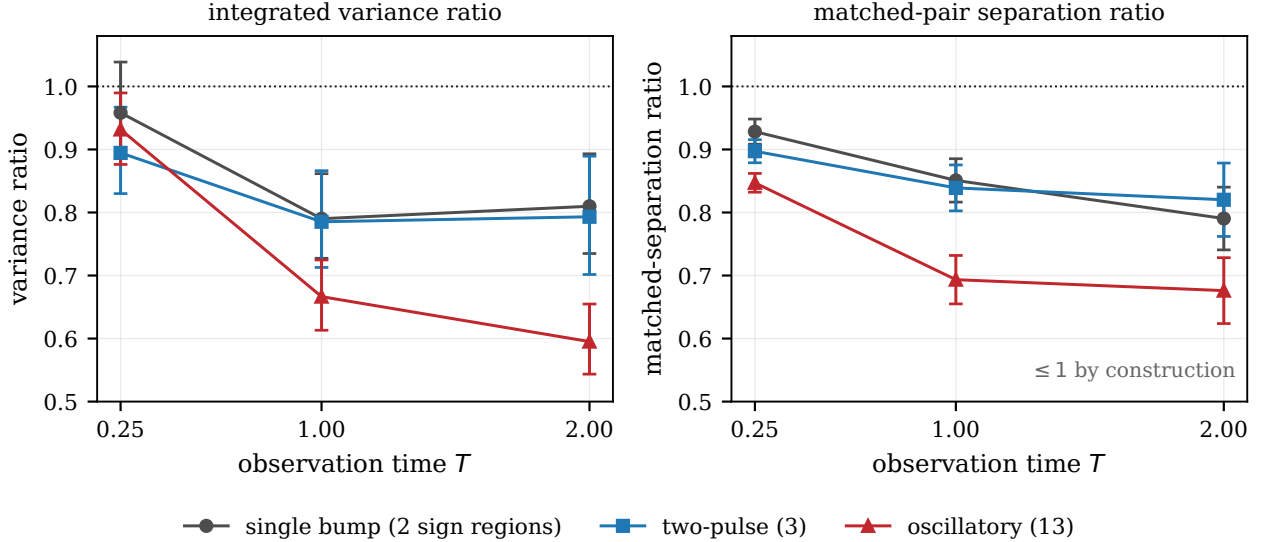


Figure 3: Left: integrated variance ratio of sign-switched to within-sign pairing across three fixed profiles and three observation times (Table 3). Right: the matched-pair separation ratio at a fixed paired state (Table 4), which cannot exceed 1 by Remark 3.9; it measures the geometric cost of constraining the matching to respect sign classes, not the variance mechanism.

establish that the slack causes the variance difference, and the two orderings are not even identical – the separation ratio at $T = 2$ orders the profiles oscillatory < single bump < two-pulse while the variance ratio orders them oscillatory < two-pulse < single bump. We therefore report the association and not a mechanism.

A separate diagnostic lets each matching rule act on its own evolved state. That comparison mixes geometry with coupling history and is recorded as a supporting association in the evidence archive; it is not used to infer a variance mechanism.

Flux. Replacing $f(u) = u^2/2$ by $f(u) = u^3/3$ changes the character of the transport: $f'(u) = u^2 \geq 0$ vanishes at $u = 0$ without changing sign, so there is no left/right propagation, and $f''(u) = 2u$ makes the characteristic speed non-affine. With everything else held fixed at $N = 1600$, $h = 0.005$ and the same 64 seeds, the sign-switched-to-within-sign variance ratio is 0.7916 [0.762, 0.825] for Burgers and 0.7688 [0.741, 0.797] for the cubic flux (Table 2). The advantage does not require bidirectional transport. This comparison does not establish transfer to arbitrary fluxes.

4.4 Work at selected accuracy targets

Variance ratios at fixed settings are not work. We ran two held-out experiments. In each, a pilot on one seed key supplied plug-in estimates of b_{hi}^2 , V and the per-replicate cost C for every arm; the replicate count of each arm was then frozen at the smallest integer B with $b_{hi}^2 + V_{hi}/B \leq \text{tol}/\kappa$, using upper endpoints and a safety factor $\kappa = 1.3$; and the frozen plans were evaluated on an unused seed key with 32 independent blocks. At each cell the paired and single arms use the switched pilot estimate of the shared bias; the RQMC arm uses its own estimate. The experiments differ in their target, their N , and their arms, and we keep them separate rather than pooling them: the first was designed before within-sign pairing was added as a comparator and does not contain it.

4.4.1 Experiment A: against an unpaired ensemble, reflected pairing, and a randomised quasi-Monte Carlo adaptation

Target 7×10^{-6} expected integrated squared field error, $N = 8192$, pilot key 1601, evaluation key 1702. The target was chosen from the pilot, so it is a pilot-informed target with an independent evaluation batch.

The quasi-Monte Carlo arm is an adaptation of L  cot’s rank-based diffusion to our transport solver, not a reproduction of his method. One `scipy.stats.qmc.Sobol` engine with $d = 2$ and 30 bits is created per replicate and scrambled once; successive time steps consume successive blocks of N points, as in his equation (45). SciPy’s `scramble=True` is a left-matrix scramble combined with a digital random shift, not fully nested Owen scrambling [24, 25]. The first coordinate assigns a rank by $\lfloor Np_1 \rfloor$ and the second supplies the increment $\sigma\Phi^{-1}(p_2)$, with p_2 clipped to $[2^{-53}, 1 - 2^{-53}]$ and the clip count recorded. N must be a power of two. We do not claim this adaptation preserves the interacting discrete law; its total error is measured independently, including its own bias. The rank-bijection assertion is checked outside the timed path.

Table 5: Held-out experiment A. Frozen plans, unused evaluation key, 32 blocks, $N = 8192$, target 7×10^{-6} . “Trajectories” counts solver runs, which is $2B$ for a paired arm and B otherwise. The upper bound is a Bonferroni-corrected 95% t bound across the four arms ($t_{31, 1-0.05/8} = 2.652$); every arm’s bound lies below the target.

arm	B	trajectories	attained error	simultaneous upper bound	measured work (s)
unpaired ensemble	65	65	5.347×10^{-6}	6.743×10^{-6}	6.658
reflected pairing	17	34	5.031×10^{-6}	6.362×10^{-6}	3.340
RQMC adaptation	46	46	4.024×10^{-6}	4.873×10^{-6}	6.848
sign-switched pairing	2	4	5.100×10^{-6}	5.611×10^{-6}	0.407

4.4.2 Experiment B: against within-sign pairing

Target 2.5×10^{-5} , $N = 2048$, pilot key 2310, evaluation key 2320, 32 blocks, tuning cost 4.05 s reported separately. This experiment uses the standard two-pulse configuration rather than the oscillatory profile that favours the construction.

Table 6: Held-out experiment B. Frozen plans, unused evaluation key, 32 blocks, $N = 2048$, target 2.5×10^{-5} , Bonferroni bound across three arms ($t_{31, 1-0.05/6} = 2.531$). Work is end-to-end with interleaved arm order.

arm	B	trajectories	attained error	simultaneous upper bound	measured work (s)
reflected pairing	18	36	1.404×10^{-5}	1.830×10^{-5}	0.615
within-sign pairing	3	6	1.769×10^{-5}	1.936×10^{-5}	0.119
sign-switched pairing	2	4	1.976×10^{-5}	2.203×10^{-5}	0.0715

All three arms attain the target, pointwise and simultaneously. The measured work ratio is 0.599 against within-sign pairing and 0.116 against reflected pairing. **Three qualifications belong with those numbers.** The arms did not land at equal attained error – reflected pairing overshoots the target most and the switched arm sits closest to it – so this is selected-plan attainment, not equal-error performance. At $B = 2$ against $B = 3$ the integer granularity carries much of the ratio: $\frac{2}{3}$ times the cost ratio 0.908 is 0.605, close to the measured 0.599, so the experiment mainly confirms that the switched arm needs one fewer replicate here. And the plans were selected from a pilot rather than optimised.

Both experiments are reported in Figure 4. For experiment B, the original evaluation collected errors with sequential arm timing. Reported work comes from a separate interleaved timing of the same frozen plans; reported errors remain those of the original held-out batch.

4.5 Observable dependence and the bias-dominated regime

4.5.1 The benefit is not uniform over the domain

At $N = 8192$ with 48 replicates on a separate seed key, the variance ratio of sign-switched to reflected pairing at four probe locations is 0.636 [0.357, 1.135] at the left flank $x = -2.0$, 0.143 [0.080, 0.255] at the left pulse $x = -0.7$, 0.093 [0.052, 0.166] at the steepest core $x = +0.1$, and 0.735 [0.412, 1.311] at the right flank $x = +1.5$ ($F(47, 47)$ intervals). The two flank comparisons are unresolved. A reader whose quantity of interest lives where the field is flat should not expect the integrated figure.

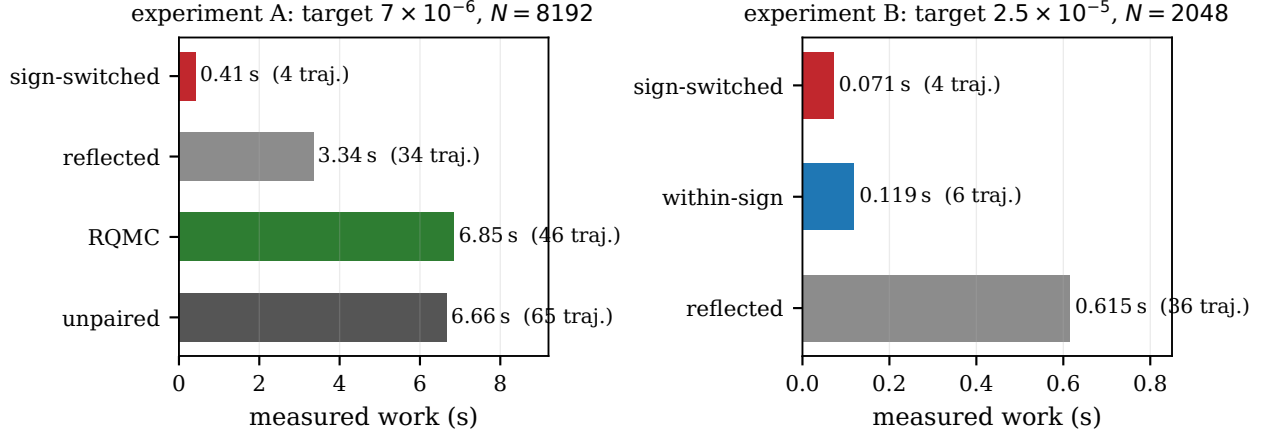


Figure 4: Measured work to attain a common expected-error target in the two held-out experiments, which have different targets, different N and different arms and are not to be pooled. Left: experiment A (Table 5). Right: experiment B (Table 6).

4.5.2 For a nonlinear functional, a variance ratio is not an accuracy ratio

Let Q be a nonlinear functional of the field. The two natural estimators, $Q(\bar{U})$ and $\frac{1}{2}(Q(U^{(A)}) + Q(U^{(B)}))$, have different expectations. By Proposition 3.1 the second has exactly the expectation of Q applied to a single uncoupled run, so its bias is the single-run bias whatever the coupling; the first has no such guarantee. We therefore report the *empirical* mean squared error against the reference,

$$\widehat{\text{MSE}} = \frac{1}{n} \sum_{r=1}^n (Q_r - Q_{\text{ref}})^2 = \widehat{\text{bias}}^2 + \frac{n-1}{n} s^2, \quad (4.2)$$

and not the plug-in $\widehat{\text{bias}}^2 + s^2$, which exceeds (4.2) by s^2/n (here 0.7%–2.0%). Table 7 uses (4.2) throughout. Two functionals are reported, and both are rule-dependent, so the rules are stated: $\max |\partial_x u|$ uses central differences of the reconstructed field on the output grid and is grid-dependent; the level crossing is the first downward crossing of $u = 0.25$ to the right of the global maximum, by linear interpolation, with replicates lacking such a crossing recorded as censored (none occurred in any cell).

Table 7: Nonlinear observables at $N = 8192$, 48 replicates, against the spectral reference, from the independently replayed archive. Ratios are of the empirical mean squared error (4.2), with whole-replicate bootstrap intervals; the last column is the squared-bias share of the sign-switched arm’s empirical MSE. An interval containing 1 is unresolved.

functional	estimator	switched / reflected	switched / within-sign	bias ² share
$\max \partial_x u $	Q of the mean	0.416 [0.244, 0.702]	0.602 [0.359, 0.984]	0.56
	mean of Q	0.832 [0.603, 1.139]	1.058 [0.803, 1.394]	0.87
level $x(u = 0.25)$	Q of the mean	0.129 [0.070, 0.247]	0.855 [0.497, 1.419]	0.19
	mean of Q	0.090 [0.053, 0.158]	0.847 [0.497, 1.486]	0.30

Table 7 contains the clearest applicability boundary in the paper. For the level crossing the switched coupling reduces total error by roughly an order of magnitude against reflected pairing under either estimator. For the peak gradient it does not: with the mean-of- Q estimator the ratio against reflected pairing is 0.832 [0.603, 1.139] and against within-sign pairing 1.058 [0.803, 1.394], both unresolved. The plug-in squared-bias share is 87% for that estimator; it is itself estimated, and Corollary 3.3 establishes sharing of the population bias across the couplings, not exact equality of finite-sample estimates. A variance reduction can only reduce the sampling component of this error. The corresponding variance ratios are 0.410 (Q of the mean) and 0.686 (mean of Q) against reflected pairing – a factor of 2.4 and 1.5, not ten.

4.5.3 The bias-dominated regime

The same arithmetic applies to the field itself. Since the couplings share b^2 , (2.9) gives

$$\frac{\mathcal{E}_{\text{SWITCH}}(B)}{\mathcal{E}_{\text{REFLECT}}(B)} = \frac{b^2 + V_{\text{SWITCH}}/B}{b^2 + V_{\text{REFLECT}}/B} \rightarrow 1 \quad \text{as } B \rightarrow \infty, \quad (4.3)$$

when $b^2 > 0$. If $b^2 = 0$, the ratio instead remains $V_{\text{SWITCH}}/V_{\text{REFLECT}}$. Thus the crossover concerns nonzero baseline bias at fixed N and h . The scale on which it fades is $B^* = V/b^2$, at which the sampling and discretisation contributions are equal. At three archived cells of the two-pulse problem B^* for the switched arm is 10.3 ($N = 6400$, $h = 0.0025$), 4.5 ($N = 6400$, $h = 0.005$) and 35.3 ($N = 1600$, $h = 0.0025$). Figure 5 uses equal continuous work instead of equal pair counts:

$$\frac{b^2 + V_{\text{SWITCH}}C_{\text{SWITCH}}/W}{b^2 + V_{\text{REFLECT}}C_{\text{REFLECT}}/W}. \quad (4.4)$$

This is a plug-in model, excluding integer allocation and timing uncertainty. The crossover marks equal sampling and bias contributions in the switched arm; it is not a cutoff for improvement.

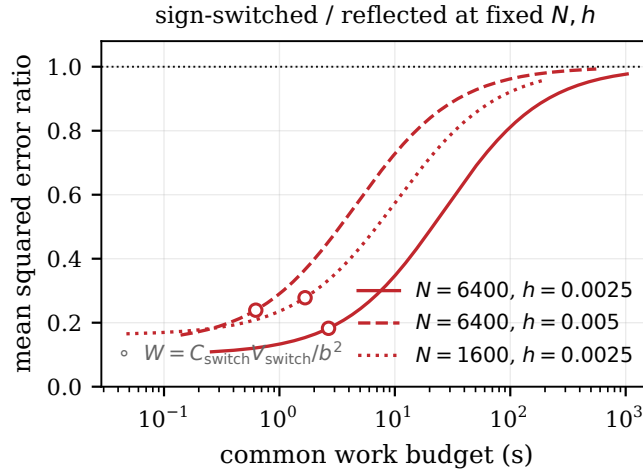


Figure 5: Model mean squared error ratio (4.4) at equal work, at three archived (N, h) cells of the two-pulse problem with 64 paired seeds. The shared squared bias b^2 is the same estimate in numerator and denominator, as Corollary 3.3 requires. Circles mark $W = C_{\text{SWITCH}}V_{\text{SWITCH}}/b^2$. Three cells indicate a direction; they do not establish a rate, and they do not separate the effects of N and h .

The practical reading is the one a reader can act on: the coupling buys accuracy when the ensemble is small enough that sampling error still matters, which is exactly the regime in which one is paying for solver trajectories. When the discretisation error dominates, refine N or h ; no coupling will help.

4.6 Signed-structure controls

Table 8 maps the effect against the signed structure of the initial data at a production cell. The rows are informative in different ways and must not be read alike.

The interleaved configuration is a mechanism test, not a refinement study. Its alternating masses $\pm 1/N$ on a fixed support give a reconstructed field whose amplitude falls like $1/N$: it is 2.5×10^{-3} at $N = 400$ and 1.2×10^{-4} at $N = 8192$, while the total absolute mass stays at 1. Increasing N therefore changes the represented field, not merely its resolution, and its ratios must not be read as convergence under refinement. The merging and separated configurations hold the amplitude at 0.5 and *are* refinements of one fixed profile.

Separated blocks. The unresolved rows are consistent with the divergence analysis of §3: the two couplings agree on about 71% of seeds and, when they do diverge, do so at a median step of 187 out of 200, leaving little history over which a difference can accumulate. This is a statement about this configuration at this N , h , ν and horizon.

Table 8: Integrated variance ratio of sign-switched to reflected pairing by signed structure, $N = 2048$, $h = 0.005$, $T = 1$, 48 replicates, whole-replicate bootstrap intervals. The two arms of each cell were run on *different* seed streams, so these are ratios of independent estimates; the one-sign row is therefore an estimate of a quantity that Corollary 3.7 makes exactly 1, and its departure from 1 is sampling noise. Exact identity is established separately, by a matched-seed full-state comparison.

structure	dynamics	variance ratio	reading
interleaved (alternating signs)	heat	0.049 [0.040, 0.061]	resolved help
	Burgers	0.063 [0.052, 0.077]	resolved help
merging blocks	heat	0.476 [0.421, 0.545]	resolved help
	Burgers	0.481 [0.426, 0.544]	resolved help
separated blocks	heat	0.994 [0.917, 1.076]	unresolved
	Burgers	1.013 [0.930, 1.101]	unresolved
one sign	heat	0.987 [0.902, 1.078]	exactly 1 by Cor. 3.7
	Burgers	0.967 [0.822, 1.174]	exactly 1 by Cor. 3.7

One sign. The two couplings are the same algorithm (Corollary 3.7); the row is a consistency check on the estimator, not evidence. The exact statement comes from a matched-seed comparison of the complete state – positions, masses and carried velocities – which gives a maximum absolute difference of exactly 0 over 200 seeds for each dynamics.

5 Discussion

The experiments distinguish two gains. Respecting mass signs through within-sign matching accounts for most of the improvement over reflected rank pairing. Retaining spatial-rank matching and switching the increment orientation gives an additional reduction, resolved in the principal field comparison and in the tested later-time profiles. The held-out experiment including within-sign pairing demonstrates a practical consequence for one selected accuracy target. Its coarse integer allocations and unequal attained errors limit interpretation as an optimized efficiency comparison.

The exact final-stage identity identifies how opposite signs, mass magnitudes, and separations affect conditional covariance. It does not characterize variance accumulated through the nonlinear histories. The separation diagnostic quantifies the geometric restriction imposed by within-sign matching; the optimality of sorted matching for absolute distance does not establish optimality for field variance. The profile comparison also changes amplitudes and gradients, so its ordering does not isolate sign-region count as a cause.

The practical benefit depends on the requested output and the balance between sampling error and baseline bias. One-sign data make the switched and reflected rules identical, while mixed-sign data can also produce little resolved gain. Nonlinear functions of the averaged field have coupling-dependent expectations. Consequently the appropriate comparison is total error for the chosen output, with the number of trajectories and execution cost reported separately.

The scope is scalar, one-dimensional mean transport with constant particle masses. The proof of marginal preservation accommodates broader predictable couplings, but neither a multidimensional matching rule nor a coupling for a sampled-velocity solver is evaluated here. All timings are serial. Parallel communication costs, finer work-to-target comparisons, and sufficient conditions for nonlinear terminal variance reduction remain open.

6 Conclusion

Sign-aware antithetic coupling combines spatial-rank matching with an increment orientation determined by the matched gradient masses. It preserves each replica’s discrete law and reduces conditional integrated variance relative to reflection when applied at the final diffusion stage. The full-history benefit is assessed numerically. In the principal two-pulse experiment, integrated variance is 0.739 times that of within-sign pairing, with approximate 95% interval [0.682, 0.801], and integrated MSE is 0.819 times as large. In a separate held-out comparison, four trajectories attain the selected target against six for within-sign pairing, at approximately 0.60 of its execution work. Fixed-profile and cubic-flux comparisons extend the empirical evidence, while one-sign controls, weak-benefit cases, and bias-dominated observables define its limits. The resulting estimator provides a simple additional option for users of signed gradient-particle solvers, with guarantees restricted to the properties proved here.

Data and code availability

The manuscript source, figure-generation scripts, numerical evidence, and reproduction instructions are distributed with the accompanying repository at <https://github.com/stephen122204/rb-gbmc-viscous-burgers>. The evidence map records source files and statistical conventions for each reported result. Historical exploratory outputs are distinguished from the curated evidence used by this manuscript.

References

- [1] Ahmed F. Ghoniem and Frederick S. Sherman. Grid-free simulation of diffusion using random walk methods. *Journal of Computational Physics*, 61(1):1–37, 1985. doi:[10.1016/0021-9991\(85\)90058-0](https://doi.org/10.1016/0021-9991(85)90058-0).
- [2] Arthur S. Sherman and Charles S. Peskin. A Monte Carlo method for scalar reaction diffusion equations. *SIAM Journal on Scientific and Statistical Computing*, 7(4):1360–1372, 1986. doi:[10.1137/0907090](https://doi.org/10.1137/0907090).
- [3] Arthur S. Sherman and Charles S. Peskin. Solving the Hodgkin–Huxley equations by a random walk method. *SIAM Journal on Scientific and Statistical Computing*, 9(1):170–190, 1988. doi:[10.1137/0909012](https://doi.org/10.1137/0909012).
- [4] Stephen Roberts. Convergence of a random walk method for the Burgers equation. *Mathematics of Computation*, 52(186):647–673, 1989. doi:[10.1090/S0025-5718-1989-0955753-4](https://doi.org/10.1090/S0025-5718-1989-0955753-4).
- [5] Arthur Sherman and Michael Mascagni. A gradient random walk method for two-dimensional reaction-diffusion equations. *SIAM Journal on Scientific Computing*, 15(6):1280–1293, 1994. doi:[10.1137/0915078](https://doi.org/10.1137/0915078).
- [6] Giulia Bertaglia, Lorenzo Pareschi, and Russel E. Caflisch. Gradient-based Monte Carlo methods for relaxation approximations of hyperbolic conservation laws. *Journal of Scientific Computing*, 100(3):Article 60, 2024. doi:[10.1007/s10915-024-02614-1](https://doi.org/10.1007/s10915-024-02614-1).
- [7] John M. Hammersley and K. William Morton. A new Monte Carlo technique: Antithetic variates. *Mathematical Proceedings of the Cambridge Philosophical Society*, 52(3):449–475, 1956.
- [8] Paul Glasserman. *Monte Carlo Methods in Financial Engineering*. Springer, New York, 2004.
- [9] Nikolas Nüsken and Grigorios A. Pavliotis. Constructing sampling schemes via coupling: Markov semigroups and optimal transport, 2018. URL <https://arxiv.org/abs/1806.11026>. arXiv:1806.11026.
- [10] Christian Lécot. Quasi-random walk methods. In *Sūrikaiseikikenkyūsho Kōkyūroku*, volume 1240, pages 103–113. Research Institute for Mathematical Sciences, Kyoto University, 2001. URL <https://www.kurims.kyoto-u.ac.jp/~kyodo/kokyuroku/contents/pdf/1240-11.pdf>.
- [11] Christian Lécot and Shigeyoshi Ogawa. Quasirandom walk methods. In Kai-Tai Fang, Harald Niederreiter, and Fred J. Hickernell, editors, *Monte Carlo and Quasi-Monte Carlo Methods 2000*, pages 63–85. Berlin, Heidelberg, 2002. Springer Berlin Heidelberg. ISBN 978-3-642-56046-0. doi:[10.1007/978-3-642-56046-0_5](https://doi.org/10.1007/978-3-642-56046-0_5).
- [12] Nadhir Ben Rached, Abdul-Lateef Haji-Ali, Raúl Tempone, and Leon Wilkosz. Forward propagation of low discrepancy through McKean–Vlasov dynamics: From QMC to MLQMC, 2024. URL <https://arxiv.org/abs/2409.09821>.
- [13] Lukas Netterdon, Veronica Montanaro, Manuel Torrilhon, and Hossein Gorji. Variance reduction in the Fokker–Planck particle method for rarefied gases using quasi-random numbers, 2026. URL <https://arxiv.org/abs/2601.14461>. arXiv:2601.14461.
- [14] Oumaima Bencheikh and Benjamin Jourdain. Bias behaviour and antithetic sampling in mean-field particle approximations of SDEs nonlinear in the sense of McKean, 2018. URL <https://arxiv.org/abs/1809.06838>. arXiv:1809.06838; published in ESAIM: Proceedings and Surveys, 2019.
- [15] Mireille Bossy and Denis Talay. A stochastic particle method for the McKean–Vlasov and the Burgers equation. *Mathematics of Computation*, 66(217):157–192, 1997. doi:[10.1090/S0025-5718-97-00776-X](https://doi.org/10.1090/S0025-5718-97-00776-X).
- [16] Mireille Bossy. Optimal rate of convergence of a stochastic particle method to solutions of 1D viscous scalar conservation laws. *Mathematics of Computation*, 73(246):777–812, 2004. doi:[10.1090/S0025-5718-03-01551-5](https://doi.org/10.1090/S0025-5718-03-01551-5).
- [17] Mireille Bossy and Benjamin Jourdain. Rate of convergence of a particle method for the solution of a 1D viscous scalar conservation law in a bounded interval. *The Annals of Probability*, 30(4):1797–1832, 2002. doi:[10.1214/aop/1039548372](https://doi.org/10.1214/aop/1039548372).
- [18] Benjamin Jourdain. Diffusion processes associated with nonlinear evolution equations for signed measures. *Methodology and Computing in Applied Probability*, 2(1):69–91, 2000. doi:[10.1023/A:1010059302049](https://doi.org/10.1023/A:1010059302049).
- [19] Oumaima Bencheikh and Benjamin Jourdain. Weak and strong error analysis for mean-field rank based particle approximations of one dimensional viscous scalar conservation law. arXiv:1910.11237v2, 2020.

- [20] Lowell L. Baker and Nicolas G. Hadjiconstantinou. Variance reduction for Monte Carlo solutions of the Boltzmann equation. *Physics of Fluids*, 17(5):051703, 2005.
- [21] Thomas M. M. Homolle and Nicolas G. Hadjiconstantinou. A low-variance deviational simulation Monte Carlo for the Boltzmann equation. *Journal of Computational Physics*, 226(2):2341–2358, 2007.
- [22] Husain A. Al-Mohssen and Nicolas G. Hadjiconstantinou. Low-variance direct Monte Carlo simulations using importance weights. *ESAIM: Mathematical Modelling and Numerical Analysis*, 44(5):1069–1083, 2010.
- [23] Cédric Villani. *Topics in Optimal Transportation*, volume 58 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2003.
- [24] Art B. Owen. Scrambling Sobol’ and Niederreiter–Xing points. *Journal of Complexity*, 14(4):466–489, 1998.
- [25] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi:[10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2).
- [26] Eberhard Hopf. The partial differential equation $u_t + uu_x = \mu u_{xx}$. *Communications on Pure and Applied Mathematics*, 3(3):201–230, 1950. doi:[10.1002/cpa.3160030302](https://doi.org/10.1002/cpa.3160030302).
- [27] Julian D. Cole. On a quasi-linear parabolic equation occurring in aerodynamics. *Quarterly of Applied Mathematics*, 9(3):225–236, 1951. doi:[10.1090/qam/42889](https://doi.org/10.1090/qam/42889).

A Proofs

A.1 Proposition 3.1

Write $\Xi_k^A = (X_k^A, m_k^A, v_k^A)$ for replica A ’s state after stage k and $\mathcal{F}_k$ for the σ -field generated by the joint history up to and including the sort and velocity update of stage k . The transport, sort and velocity update of Algorithm 1 are deterministic functions of Ξ_{k-1}^A alone. It therefore suffices to show that, conditionally on $\mathcal{F}_k$, the vector of increments applied to replica A is $\mathcal{N}(0, \sigma^2 I_N)$, and likewise for B .

For A this is immediate: $Z^{(k)} \sim \mathcal{N}(0, I_N)$ is drawn independently of $\mathcal{F}_k$. For B , the applied increment is $\sigma \varepsilon_{k,j} Z_{\pi_k(j)}^{(k)}$ with $(\pi_k, \varepsilon_k) \mathcal{F}_k$ -measurable. Conditionally on $\mathcal{F}_k$ the pair (π_k, ε_k) is a fixed bijection and a fixed sign vector, while $Z^{(k)}$ remains $\mathcal{N}(0, I_N)$; a permutation of the coordinates of an i.i.d. standard Gaussian vector is i.i.d. standard Gaussian, and multiplying coordinate j by a constant of modulus one leaves its law $\mathcal{N}(0, 1)$ and preserves independence across j because the permutation is injective. Hence replica B ’s conditional increment law is $\mathcal{N}(0, \sigma^2 I_N)$, the same as an uncoupled run’s. Induction over $k = 1, \dots, K$ gives equality of the laws of the complete state trajectories. Only the joint law of (Ξ^A, Ξ^B) is changed, since the increments of the two replicas are now dependent. $\square$

Each hypothesis is used: freshness and independence of $Z^{(k)}$ from $\mathcal{F}_k$ for the conditional law; bijectivity of π_k for independence across coordinates; unit modulus for the variance; measurability with respect to $\mathcal{F}_k$ so that the multipliers are not chosen after seeing the draw; exclusion of zero masses so that (2.4) has modulus one; and retention of the particle–velocity association until transport so that Ξ is the complete state.

A.2 Lemma 3.4

Fix $c \in \mathbb{R}$ and set $h_a(x) = \mathbf{1}\{x \geq a\} - \mathbf{1}\{x \geq c\}$, which is compactly supported with $\int h_a h_b dx = \frac{1}{2}(|a - c| + |b - c| - |a - b|)$ by direct computation. Since $\sum_i m_i^A = \sum_j m_j^B = 0$ we may write $D_A = \sum_i m_i^A h_{X_i^A}$ and $D_B = \sum_j m_j^B h_{X_j^B}$, both compactly supported. Then

$$\int D_A D_B = \sum_{i,j} m_i^A m_j^B \frac{1}{2} (|X_i^A - c| + |X_j^B - c| - |X_i^A - X_j^B|),$$

and the first two groups of terms vanish because $\sum_j m_j^B = 0$ and $\sum_i m_i^A = 0$ respectively. $\square$

A.3 Proposition 3.5

Condition on the pre-final state. Write $D_\bullet$ as in Lemma 3.4 for the terminal configurations, so that $\bar{U} = u_- + \frac{1}{2}(D_A + D_B)$ and

$$V_\bullet = \int \text{Var } \bar{U} = \frac{1}{4} \left(\int \text{Var } D_A + \int \text{Var } D_B + 2 \int \text{Cov}(D_A, D_B) \right).$$

By Proposition 3.1 the first two integrals do not depend on the coupling. For the third, Lemma 3.4 applied to $\mathbb{E} \int D_A D_B$ and, separately, to $\int \mathbb{E} D_A \mathbb{E} D_B$ – the latter by introducing an independent copy Z' of the noise and using $\mathbb{E} D_A(x) = \mathbb{E} Z[\cdot]$ – gives

$$\int \text{Cov}(D_A, D_B) = -\frac{1}{2} \sum_{i,j} m_i^A m_j^B \left[\mathbb{E} |X_i^A - X_j^B| - g_{\sigma\sqrt{2}}(a_i - b_j) \right].$$

For $i \neq j$ the increments Z_i and $\varepsilon_j Z_j$ are independent with variance σ^2 each under REFLECT and SWITCH, so $X_i^A - X_j^B = (a_i - b_j) + \sigma\sqrt{2}\mathcal{N}(0, 1)$ in law and the bracket vanishes. Only the diagonal survives. There, under REFLECT, $X_i^A - X_i^B = d_i + 2\sigma Z_i$ and the first term is $g_{2\sigma}(d_i)$; under SWITCH it is $g_{2\sigma}(d_i)$ when $m_i^A m_i^B > 0$ and, when $m_i^A m_i^B < 0$, the two increments cancel exactly and the first term is $|d_i|$. Subtracting,

$$\int \text{Cov}_{\text{REFLECT}} - \int \text{Cov}_{\text{SWITCH}} = -\frac{1}{2} \sum_{m_i^A m_i^B < 0} m_i^A m_i^B [g_{2\sigma}(d_i) - |d_i|] = \frac{1}{2} \sum_{m_i^A m_i^B < 0} |m_i^A m_i^B| [g_{2\sigma}(d_i) - |d_i|],$$

and multiplying the covariance difference by $1/2$ gives (3.2). Non-negativity is $g_s(d) = \mathbb{E}|d + sZ| \geq |\mathbb{E}(d + sZ)| = |d|$. $\square$

B A two-particle monotonicity counterexample

Take $u_- = 0$, $m = (1/2, -1/2)$, $h = 0.005$, $\nu = 10^{-10}$, and two steps. The initial positions are $(p, 0)$, with $p_L = -0.00275$ or $p_R = -0.00225$. Both initial carried velocities are $(1/2, 0)$. Without diffusion, the first transport puts the positive particle at -0.00025 in state L and $+0.00025$ in state R . Re-sorting therefore leaves the velocities $(1/2, 0)$ in L but assigns $(-1/2, 0)$ to the negative and positive particles, respectively, in R . After the second transport, the particle positions in identity order are

$$L : (0.00225, 0), \quad R : (0.00025, -0.0025).$$

At the probe $x_* = 0.00125$, the reconstructed fields are $-1/2$ and 0 .

Now restore the Gaussian displacements with $\sigma = 10^{-6}$. On the event that each of the four displacements of a trajectory has modulus at most $\delta = 10^{-4}$, neither its second transport ordering nor its terminal probe classification changes: the relevant ordering gap is $0.00025 > 2\delta$ and the smallest probe margin is $0.001 > 2\delta$. Hence the same field values hold on this event. For either initial state the exceptional probability is at most $\epsilon = 8\Phi(-100) < \exp(-5003)$. Since every field value is in $[-1/2, 1/2]$,

$$\mathbb{E} U_R(x_*, 2h) - \mathbb{E} U_L(x_*, 2h) \geq \frac{1}{2} - \frac{3}{2}\epsilon > 0.$$

The initial movement of a positive mass to the right thus increases the expected terminal field. This proves the claim in Remark 3.8 without inferring an expectation from a finite set of simulation seeds.

C Test problems, initialisation and reference solutions

Initialisation. For the smooth profiles with $u_+ = u_-$, let $\text{TV} = \int |u'_0|$. We place $N/2$ particles at the deterministic midpoint quantiles $r_k = (k - \frac{1}{2})/(N/2)$, $k = 1, \dots, N/2$, of the normalised positive part of u'_0 and $N/2$ at those of the negative part, assigning $m_i = \pm \text{TV}/N$. This gives equal counts and equal absolute mass in each sign class and makes $\sum_i m_i = u_+ - u_-$; it is deterministic, so the initial state is common to both replicas and to every arm.

Profiles. The two-pulse profile is (4.1). The single-bump profile is $0.8 e^{-(x+0.2)^2/(2 \cdot 0.45^2)}$ and the oscillatory profile is $0.45 \sin(3x) e^{-x^2/(2 \cdot 1.1^2)}$; the number of sign regions of u'_0 quoted for each is computed from a 40 001-point evaluation with a relative magnitude threshold of 10^{-6} , giving 2, 3 and 13. The four signed-structure configurations of §4.6 place N particles of equal magnitude $1/N$ as follows: *interleaved*, alternating signs on $[-0.25, 0.25]$; *separated*, a positive block on $[-2.0, -1.4]$ and a negative block on $[1.4, 2.0]$; *merging*, blocks on $[-0.9, -0.5]$ and $[0.5, 0.9]$; *one sign*, all masses $-2/N$ on $[-1, 1]$.

References. For $f(u) = u^2/2$ and $f(u) = u^3/3$ the reference is a Fourier pseudospectral solution on $[-16, 16]$ with $M = 2048$ modes, $\frac{2}{3}$ -rule dealiasing and integrating-factor RK4 at $dt = 10^{-4}$. For Burgers it agrees with a stabilised whole-line Cole–Hopf evaluation [26, 27] – the potential $F(y) = \int_0^y u_0$ being analytic for Gaussian pulses – to 5.57×10^{-14} in the maximum norm on the output grid; for the cubic flux, halving dt and doubling M moves the solution by 1.6×10^{-13} . These are correctness instruments. They are never used as cost comparators and never enter the fitting of any control coefficient.

D Uncertainty methodology and its calibration

The two interval procedures used for scalar observables are a whole-replicate percentile bootstrap and $F(n-1, n-1)$. The second assumes independent normal replicates, which we do not assert, so we calibrated both.

Per-replicate values of four observables were taken from the archived $N = 8192$ fields, *separately* for the reflected and sign-switched arms, and each sample was centred and scaled by its own standard deviation to give two empirical shapes. A synthetic numerator arm resamples the sign-switched shape, scaled to impose a known variance ratio r ; the denominator arm resamples the reflected shape. Nominal-95% intervals are then formed by each procedure and coverage counted over $M = 1500$ synthetic datasets, with $n = 48$ per arm and 400 bootstrap resamples per interval. Table 9 reports the two-shape design together with the common-shape design of an earlier calibration, which used one standardised sample for both arms and therefore could not speak to the case of interest.

Table 9: Coverage of nominal-95% variance-ratio intervals, $M = 1500$ synthetic datasets per cell, binomial standard error ≤ 0.007 . “Two-shape” resamples the sign-switched and reflected empirical shapes separately; “common-shape” reproduces the earlier design in which both arms came from one standardised sample.

observable	two-shape, $r = 0.15$		two-shape, $r = 0.75$		common-shape, $r = 0.75$	
	bootstrap	F	bootstrap	F	bootstrap	F
station $x = +0.1$	0.921	0.958	0.933	0.967	0.945	0.977
station $x = -2.0$	0.950	0.982	0.946	0.973	0.942	0.977
$\max \partial_x u $	0.941	0.965	0.931	0.960	0.947	0.954
level $x(u = 0.25)$	0.933	0.959	0.928	0.949	0.935	0.963

Under the two-shape design, estimated F coverage is 0.949–0.982 and percentile-bootstrap coverage is 0.921–0.950, with simulation uncertainty as stated in the table. We therefore report F intervals for station and level-crossing variance ratios and bootstrap intervals for integrated, and mean-squared-error ratios. The calibration concerns scalar variance ratios; it does not establish coverage for integrated-field or MSE intervals. Those bootstrap intervals are approximate, particularly at these sample sizes. This describes performance under *these* empirical distributions at *this* configuration; it is not a guarantee for other problems, and a non-significant normality test records failure to reject, not normality.

E Evidence provenance

Table 10 maps every table and figure to the archived output that produced it. Paths are relative to the study repository’s output/ directory unless stated otherwise; each directory name carries a date suffix (`_2026_09_09` or `_2026_09_10`) which is omitted here for width. The file `analysis/numbers.json` contains each quantity together with the archive it was read from.

Table 10: Provenance of reported results.

item	what it answers	archive
Table 1, Fig. 2	headline field comparison	independent replay of round22_corrections; replayed_fields.npz, replay_audit.json
cost figures, §4.2	what the coupling costs	round22_corrections/costs.json; round17_compare/compare.json (pair versus single)
Table 2	against controls and a second flux	round09_costs_control/fields.npz, .../rows.csv
Table 3, Fig. 3 (left)	profiles and times	round23_mechanism/mechanism.json
Table 4, Fig. 3 (right)	matched-pair geometry	round24_separation/separation.json (matched_state rows; the own_history rows are the association-only variant, one of nine cells containing 1)
Table 5, Fig. 4 (left)	held-out work, experiment A	round17_compare/compare.json
Table 6, Fig. 4 (right)	held-out work, experiment B	round23_mechanism/worktarget.json; interleaved timings from round24_separation/timing_check.json
station ratios, §4.5	where the benefit sits	round18_observable/observable.json, .../f_intervals.json
Table 7	nonlinear observables	replayed archive as above; empirical MSE by (4.2)
Fig. 5	bias-dominated regime	round07_joint_cell/joint_cell.json
Table 8	signed-structure map	round21_applicability/applicability.json
Prop. 3.5 check and share	is the identity right, and how much does it explain	round25_finalstage/finalstage.json; aggregates from round06_finalstep/finalstep.json
Cor. 3.7, divergence fractions	when are the couplings identical	round25_divergence/divergence.json (200 seeds); round22_corrections/corrections.json (40 seeds)
Prop. 3.1 witness	does a replica match production	round14_audit/audit.json; single_vs_production maximum absolute differences 0 in position, mass and field
Rem. 3.8	why there is no unconditional guarantee	round07_counterexample/counterexample.json
Table 9	are the intervals trustworthy	round25_calibration/calibration.json (supersedes the single-shape design in round22_corrections/calibration.json)